WMDR2-core

Package in package 'Model'

WMDR2-core
Version 0.1 Phase 1.0 Proposed
jkl created on 2026-03-06. Last modified 2026-03-11

WMDR2-core diagram

Class diagram in package 'WMDR2-core'

WMDR2core
Notes
1. temporalExtent shall be realized as time:interval
2. temporal<Name> elements other than temporalExtent shall be realized as an array of two arrays, one for times, one for the values of <Name>, e.g., temporalTerritory = [[2000-01-01, 2024-07-13], ["https://codes.wmo.int/wmdr/territory/<ISO1>", "https://codes.wmo.int/wmdr/territory/<ISO2>"]].
3. Following the OGC Features model, geometry = geolocation = coordinates + elevation

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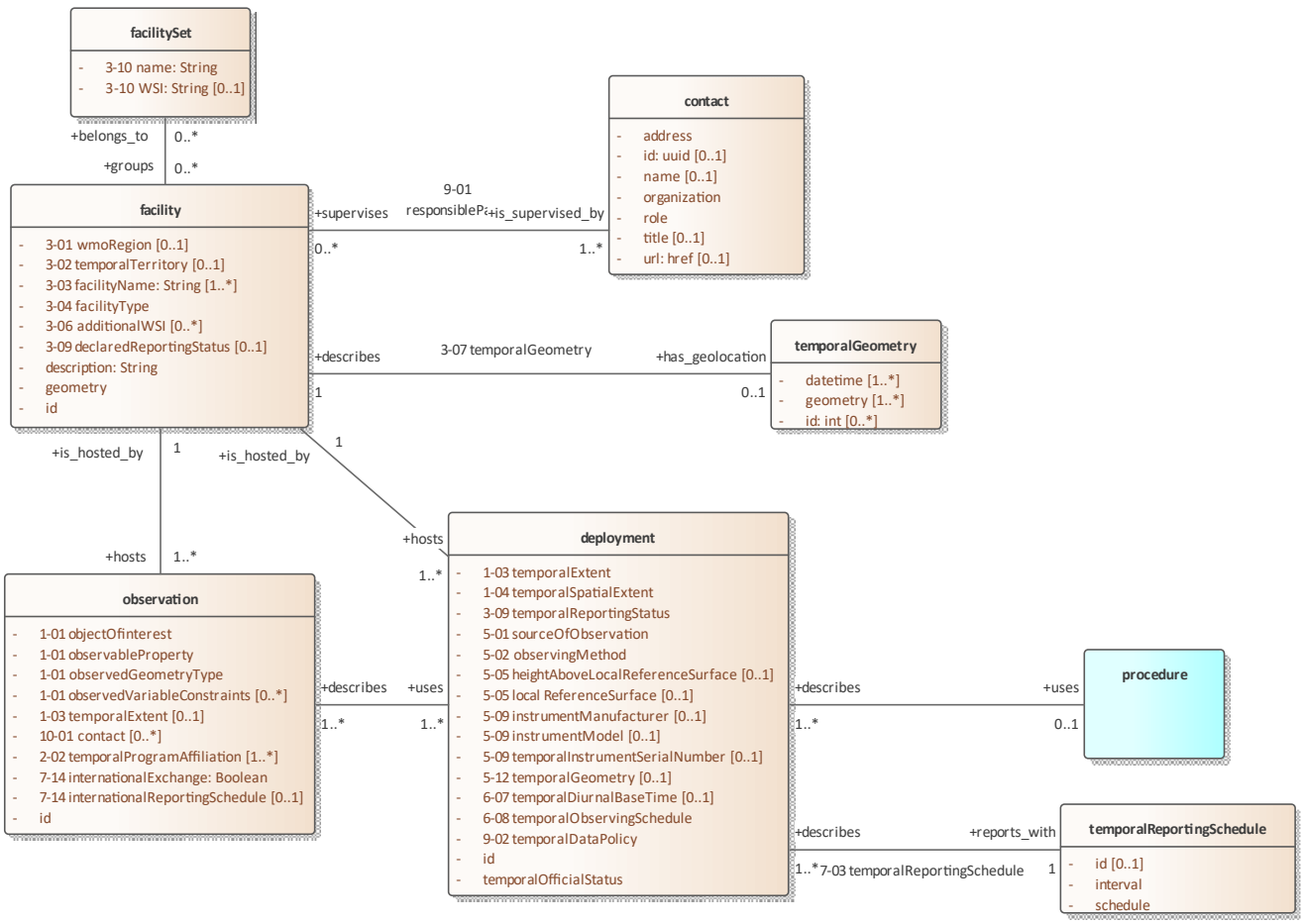

Figure 1: WMDR2-core

contact

Class in package 'WMDR2-core'

impkement according to VCard 4.0 specification [0].

[0] https://datatracker.ietf.org/doc/html/rfc6350#page-23

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ATTRIBUTES

address : Private

ADR: The structured type value consists of a sequence of
address components. The component values MUST be specified in
their corresponding position. The structured type value
corresponds, in sequence, to
the post office box;
the extended address (e.g., apartment or suite number);
the street address;
the locality (e.g., city);
the region (e.g., state or province);
the postal code;
the country name (full name in the language specified in
Section 5.1).

Example:
ADR;TYPE=work:;;Federal Department of Home Affairs FDHA Federal Office of M
eteorology and Climatology MeteoSwiss Measurements and Data\nOperation Cen
ter 1 (OPC 1)\nP.O. Box 257;Zürich-Flughafen;;CH-8058;Switzerland

[Is static True. Containment is Not Specified.]

id : uuid Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specified.]

name : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Either a full name, or a sequence of name parts. Examples:

Dr. Jörg Klausen (this corresponds to FN in vcard)
Klausen;Jörg;Dr.; (this corresponds to N in vcard)

[Is static True. Containment is Not Specified.]

organization : Private

Example:
ORG:Federal Office of Meteorology and Climatology MeteoSwiss

[Is static True. Containment is Not Specified.]

role : Private

[Is static True. Containment is Not Specified.]

ATTRIBUTES

title : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
[Is static True. Containment is Not Specified.]

url : href Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified) 9-01 responsibleParty

Source: Public supervises (Class) facility
Cardinality: [0..*]

Target: Public is_supervised_by (Class) contact
Cardinality: [1..*]

deployment

Class in package 'WMDR2-core'

deployment
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ATTRIBUTES

1-03 temporalExtent : Private
Alias: records:interval
[Is static True. Containment is Not Specified.]

1-04 temporalSpatialExtent : Private
Alias: records:geometry

Typical spatial geo-referenced volume covered by the observations.

EXAMPLES:
(a) Air temperature at a surface observing site: Sydney Airport, New South Wales, Australia: lat. 33.9465°N, lon. 151.1731°E, alt. 6.0 m above mean sea level (amsl);
(b) The projected area or volume of the cone around a particular weather radar with a maximum range of 370 km (radar reflectivity) and 150 km (Doppler); to be expressed as a geometric shape;
(c) Three-dimensional grid of radar pixels;
(d) For infrared and visible imagery by meteorological satellite (sun-synchronous): global;
(e) For nadir sounding: atmospheric column above ocean;
(f) The Japan Meteorological Agency Advanced Himawari Imager has a 10 min cycle full disk scan (0.5-2.0 km spatial resolution) located above 140.68°E geostationary orbit at an altitude of 35,786 km;
(g) River discharge by gauge: size and geometric shape of a river catchment.

Q: what is the spatial extent for a)?
[Is static True. Containment is Not Specified.]

3-09 temporalReportingStatus : Private
[Is static True. Containment is Not Specified.]

ATTRIBUTES

5-01 sourceOfObservation : Private

[Is static True. Containment is Not Specified.]

5-02 observingMethod : Private

[Is static True. Containment is Not Specified.]

5-05 heightAboveLocalReferenceSurface : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

implement as array

[Is static True. Containment is Not Specified.]

5-05 local ReferenceSurface : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Linked to heightAboveReferenceSurface

[Is static True. Containment is Not Specified.]

5-09 instrumentManufacturer : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Change of instrument requires a new deployment

[Is static True. Containment is Not Specified.]

5-09 instrumentModel : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Change of instrument requires a new deployment

[Is static True. Containment is Not Specified.]

5-09 temporalInstrumentSerialNumber : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specified.]

5-12 temporalGeometry : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

- Geospatial location of instrument/sensor

[Is static True. Containment is Not Specified.]

6-07 temporalDiurnalBaseTime : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

indicates the end (hour) of a 24-hour period. Mandatory for 24-hour periodic observations, e.g., cumulative amount of precipitation

[Is static True. Containment is Not Specified.]

6-08 temporalObservingSchedule : Private

complex type: when are observations made? Should be expressed as a temporal coverage, e.g. "from Feb-Oct on working days

ATTRIBUTES

from 06-18 UTC"

[Is static True. Containment is Not Specified.]

9-02 temporalDataPolicy : Private

remove from WMDR2 and leave for WCMP2?

[Is static True. Containment is Not Specified.]

id : Private

[Is static True. Containment is Not Specified.]

temporalOfficialStatus : Private

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public describes (Class) deployment
Cardinality: [1..*]

Target: Public uses (Class) procedure
Cardinality: [0..1]

ObservingProcedure(s)

Association (direction: Unspecified) 7-03 temporalReportingSchedule

Source: Public describes (Class) deployment
Cardinality: [1..*]

Target: Public reports_with (Class) temporalReportingSchedule
Cardinality: [1]

Association (direction: Unspecified) 5-09 instrument

Source: Public describes (Class) deployment
Cardinality: [0..*]

Target: Public uses (Class) instrument
Cardinality: [0..1]

Association (direction: Unspecified)

Source: Public describes (Class) observation
Cardinality: [1..*]

Target: Public uses (Class) deployment
Cardinality: [1..*]

Association (direction: Unspecified)

Source: Public is_hosted_by (Class) facility
Cardinality: [1]

Target: Public hosts (Class) deployment
Cardinality: [1..*]

facility

Class in package 'WMDR2-core'

facility
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ATTRIBUTES

3-01 wmoRegion : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

mandatory for fixed facilities, optional for mobile facilities.

[Is static True. Containment is Not Specified.]

3-02 temporalTerritory : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

mandatory for fixed facilities, optional for mobile facilities

[Is static True. Containment is Not Specified.]

3-03 facilityName : String Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specified.]

3-04 facilityType : Private

[Is static True. Containment is Not Specified.]

3-06 additionalWSI : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

An array of well-formed WSIs

[Is static True. Containment is Not Specified.]

3-09 declaredReportingStatus : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Condition: must be provided if no deployment is specified; otherwise: Optional, inferred as "best case" from deployments: temporalReportingStatus + temporalProgramAffiliation.
Exception: "closed" over-rules deployment, in which case all deployments should be terminated.

[Is static True. Containment is Not Specified.]

description : String Private

[Is static True. Containment is Not Specified.]

geometry : Private
Alias: 3-07 current geolocation

geometry = geolocation = coordinates + elevation

[Is static True. Containment is Not Specified.]

id : Private
Alias: 3-06 primaryWSI

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public is_hosted_by (Class) facility
Cardinality: [1]

Target: Public hosts (Class) observation
Cardinality: [1..*]

Association (direction: Unspecified) 3-07 temporalGeometry

Source: Public describes (Class) facility
Cardinality: [1]

Target: Public has_geolocation (Class) temporalGeometry
Cardinality: [0..1]

Association (direction: Unspecified) 9-01 responsibleParty

Source: Public supervises (Class) facility
Cardinality: [0..*]

Target: Public is_supervised_by (Class) contact
Cardinality: [1..*]

Association (direction: Unspecified)

Source: Public (Class) facility

Target: Public (InformationItem) temporalGeometry

Association (direction: Unspecified)

Source: Public groups (Class) facility
Cardinality: [0..*]

Target: Public belongs_to (Class) facilitySet
Cardinality: [0..*]

Association (direction: Unspecified)

Source: Public is_hosted_by (Class) facility
Cardinality: [1]

Target: Public hosts (Class) deployment
Cardinality: [1..*]

facilitySet

Class in package 'WMDR2-core'

facilitySet
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ATTRIBUTES

3-10 name : String Private

[Is static True. Containment is Not Specified.]

3-10 WSI : String Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Can facilitySet have a WSI? If not, this can be dropped.

[Is static True. Containment is Not Specified.]

ATTRIBUTES

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public groups (Class) facility
Cardinality: [0..*]

Target: Public belongs_to (Class) facilitySet
Cardinality: [0..*]

instrument

Class in package 'WMDR2-core'

instrument
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ATTRIBUTES

id : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
[Is static True. Containment is Not Specified.]

manufacturer : Private

This should come from a registry/catalogue, but should accept any organization.
[Is static True. Containment is Not Specified.]

model : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

This should come from a registry/catalogue, but can accept "custom-built".
[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified) 5-09 instrument

Source: Public describes (Class) deployment
Cardinality: [0..*]

Target: Public uses (Class) instrument
Cardinality: [0..1]

observation

Class in package 'WMDR2-core'

observation
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ATTRIBUTES

1-01 objectOfinterest : Private

observedVariable should be modeled using the I-ADOPT ontology approach. Combination of
- observableProperty: a state variable (e.g., temperature), or another property (e.g., concentration of something)
- objectOfInterest: e.g., dry air (of the atmosphere)
- observedVariableConstraint(s): e.g., a size fraction, a specific wavelength, a pressure (range), a temperature (range), a humidity (range), etc., as relevant for the result of the observation. Constraints shall be documented as name-value pairs, where the names shall come from a code list, and the values shall come from a code list depending on the name, or be free text.

[Is static True. Containment is Not Specified.]

1-01 observableProperty : Private

observedVariable should be modeled using the I-ADOPT ontology approach. Combination of
- observableProperty: a state variable (e.g., temperature), or another property (e.g., concentration of something)
- objectOfInterest: e.g., dry air (of the atmosphere)
- observedVariableConstraint(s): e.g., a size fraction, a specific wavelength, a pressure (range), a temperature (range), a humidity (range), etc., as relevant for the result of the observation. Constraints shall be documented as name-value pairs, where the names shall come from a code list, and the values shall come from a code list depending on the name, or be free text.

[Is static True. Containment is Not Specified.]

1-01 observedGeometryType : Private

[Is static True. Containment is Not Specified.]

1-01 observedVariableConstraints : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

observedVariable should be modeled using the I-ADOPT ontology approach. Combination of
- observableProperty: a state variable (e.g., temperature), or another property (e.g., concentration of something)
- objectOfInterest: e.g., dry air (of the atmosphere)
- observedVariableConstraint(s): e.g., a size fraction, a specific wavelength, a pressure (range), a temperature (range), a humidity (range), etc., as relevant for the result of the observation. Constraints shall be documented as name-value pairs, where the names shall come from a code list, and the values shall come from a code list depending on the name, or be free text.

[Is static True. Containment is Not Specified.]

1-03 temporalExtent : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

aggregate derived from deployments if deployments are provided; otherwise, specify begin-end

[Is static True. Containment is Not Specified.]

10-01 contact : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

ogc:contact includes "role"

[Is static True. Containment is Not Specified.]

2-02 temporalProgramAffiliation : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specified.]

ATTRIBUTES

7-14 internationalExchange : Boolean Private

[Is static True. Containment is Not Specified.]

7-14 internationalReportingSchedule : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

complex type: when are observations reported internationally? Should be expressed as a temporal coverage

Conditional: report if internationalExchange==True

[Is static True. Containment is Not Specified.]

id : Private

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public describes (Class) observation
Cardinality: [1..*]

Target: Public uses (Class) deployment
Cardinality: [1..*]

Association (direction: Unspecified)

Source: Public is_hosted_by (Class) facility
Cardinality: [1]

Target: Public hosts (Class) observation
Cardinality: [1..*]

procedure

Class in package 'WMDR2-core'

Extension point to document
- ObservingProcedure
- (Raw)DataProcessing
- ...

procedure
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ASSOCIATIONS

Association (direction: Unspecified)

Source: Public describes (Class) deployment
Cardinality: [1..*]

Target: Public uses (Class) procedure
Cardinality: [0..1]

ObservingProcedure(s)

temporalGeometry

Class in package 'WMDR2-core'

temporalGeometry
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ATTRIBUTES

datetime : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)
[Is static True. Containment is Not Specified.]

geometry : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)
[Is static True. Containment is Not Specified.]

id : int Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)
[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified) 3-07 temporalGeometry

Source: Public describes (Class) facility
Cardinality: [1]

Target: Public has_geolocation (Class) temporalGeometry
Cardinality: [0..1]

temporalReportingSchedule

Class in package 'WMDR2-core'

temporalReportingSchedule
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ATTRIBUTES

id : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
[Is static True. Containment is Not Specified.]

interval : Private

An array of arrays of begin(inclusive) and end(exclusive) datetime stamps. If the end is not specified, this indicates open-ended. Intervals can overlap to indicate multiple schedules operating in parallel.
[Is static True. Containment is Not Specified.]

schedule : Private

An array of arrays, whereby each nested array corresponds to an interval and describes a schedule. Schedules are specified as follows:

ATTRIBUTES

...

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified) 7-03 temporalReportingSchedule

Source: Public describes (Class) deployment
Cardinality: [1..*]

Target: Public reports_with (Class) temporalReportingSchedule
Cardinality: [1]

temporalGeometry

InformationItem in package 'WMDR2-core'

temporalGeometry
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ASSOCIATIONS

Association (direction: Unspecified)

Source: Public (Class) facility

Target: Public (InformationItem) temporalGeometry